C 62703 (Pages : 4) Name.....

Reg. No.....

SECOND SEMESTER B.Sc. DEGREE EXAMINATION, MAY 2014

(U.G.—CCSS)

Complementary Course

MM 2C 02—MATHEMATICS

Time : Three Hours Maximum : 30 Weightage

Unit I

Answer all **twelve** *questions.*
Each question carries ¼ weightage.

1. Tanh x is equal to :

   (a) $\frac{e^x - e^{-x}}{2}$. (b) $\frac{e^x + e^{-x}}{2}$.

   (c) $\frac{e^x - e^{-x}}{e^x + e^{-x}}$. (d) $\frac{e^x + e^{-x}}{e^x - e^{-x}}$.

2. Write $csch^{-1}x$ as a logarithmic function :

3. Find $\frac{d}{dx}\left(6\sinh\frac{x}{3}\right)$.

4. Find $\int_0^1 \frac{1}{\sqrt{x}}\,dx$.

5. The nth term of the sequence

   0, 3, 8, 15, 24 .....

6. Find $\lim_{n\to\infty} \sqrt[n]{3n}$.

7. Find the sum of the series $\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}}$.

8. Test the convergence of the series

   $1 + \frac{1}{3} + \frac{1}{7} + \frac{1}{15} + ...$

9. Define the absolute convergence of a series Σa_n.

10. Find the Cartesian equation of the curve $r = 6\sin\theta$.

11. Write the polar equation of the hyperbola with $k = 2$ and $e = \frac{3}{2}$.

12. $f(x, y) = 100 - x^2 - y^2$. Find the level curve of $f(x, y) = 75$.

(12 × ¼ = 3 weightage)

Unit II

Answer any **nine** *questions.*
Each question carries 1 *weightage.*

13. Differentiate $\tanh\sqrt{1+t^2}$ with respect to t.

14. Find $\int \operatorname{sech}^2\left(x - \frac{1}{2}\right) dx$.

15. Find $\int_0^2 \frac{dx}{1-x^2} dx$.

16. Find the sum of the series $\sum_{n=1}^{\infty} \frac{6}{(2n-1)(2n)}$.

17. Does the series $1 + \frac{1}{3} + \frac{1}{7} + \frac{1}{15} + \ldots$. converge ?

18. Show that the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ is absolutely convergent.

19. For what value of x does the power series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$ converge ?

20. Find the Taylor series expansion of $f(x) = \ln(1+x)$ at $x = 0$.

21. Find the polar equation of the elliptic $4x^2 + 9y^2 = 36$.

22. Write the centre and radius of the circle $r + \cos\theta = 0$.

23. Find $\lim_{p \to (0,-2,0)} \left(\ln \sqrt{x^2 + y^2 + z^2}\right)$.

24. Find $\frac{\partial f}{\partial x}$ at $(4, -5)$ if $f(x, y) = x^2 + 3xy + y$.

(9 × 1 = 9 weightage)

Unit III

Answer any **five** *questions.*
Each question carries 2 *weightage .*

25. Show that $\cosh^{-1} x = \ln\left(x + \sqrt{x^2 - 1}\right)$, $x \geq 1$.

26. Find $\int_{-1}^{\infty} \frac{d\theta}{\theta^2 + 5\theta + 6}$.

27. Show that $\frac{1 + 2\ln 2}{9} + \frac{1 + 3\ln 3}{14} + \frac{1 + 4\ln 4}{2} + \ldots\ldots$ diverges.

28. Find the points of intersection of the pair of curves $r = 1 + \cos\theta$ and $r = 1 - \cos\theta$.

29. If $f(x, y) = x\cos y + y\,e^x$ find $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y^2}$.

30. Express $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ interms of r and s where $w = x^2 + y^2$, $x = r + s$, $y = r - s$.

31. Find the directional derivative of $f(x, y) = 3xy - y^2$ at $(5, 5)$ with directions of $\bar{A} = 4i + 3j$.

32. Find the area that lies inside the circle $r = 1$ and outside $r = 1 - \cos\theta$.

(5 × 2 = 10 weightage)

Unit IV

Answer any **two** *questions.*
Each question carries 4 *weightage.*

33. (a) Evaluate $\int_2^{\infty} \frac{x+3}{(x-1)(x^2+1)}\,dx$.

(b) Investigate the convergence of $\sum_{n=1}^{\infty} \frac{4^n (n!)^2}{(2n)!}$.

34. Find the length of the cardioid $r = 1 - \cos\theta$.

35. Find the linearization of $f(x, y) = x^2 - xy + \frac{y^2}{2} + 3$ at the point (3, 2).

(2 × 4 = 8 weightage)